

SOLUTIONS Exam 2 (November 9/13)

QTM 220

Problem 1

Suppose we have a sample of $Y_1 \dots Y_n$ and $X_1 \dots X_n$ stored in an **R** dataframe ‘df’ with n rows. X can take the values of 1,2,3,4 or 5. We run this code.

```
B=10000
theta<- rep(NA,B)
for( b in 1:B ){
  bootind = sample(1:n,size=n,replace=T)
  dfboot = df[bootind,]
  theta[b] = (1/4)*(mean(dfboot$y[dfboot$x==5]) - mean(dfboot$y[dfboot$x==1]))
}
```

We have a **sample** of data Y and X , which take the values 1 through 5. What is this code doing?

Sample with replacement from our dataframe, calculate the average incremental difference in Y across values of X for each sample, save that estimate in a vector called theta.

How do we know this is the average incremental difference estimator? This is the same as our “average over years” for education, where we used

$$\frac{1}{4} \sum_{x=2}^5 [\hat{\mu}(x) - \hat{\mu}(x-1)] = \frac{1}{4} (\hat{\mu}(5) - \hat{\mu}(1))$$

The only difference is our X values went from 8 to 12 instead of 1 to 5.

If it was just a difference in means, it would be $\hat{\mu}(5) - \hat{\mu}(1)$ (without the 1/4). If it was weighted by the number of observations in each level of X , we would have to estimate some proportion P_X in each bootstrap iteration.

So what is theta? It is our vector that stores the incremental difference estimate for each bootstrap replication. This is the estimated sampling distribution of our incremental difference estimator. Note theta does not store a sample– it is the estimate we calculated from any individual sample.

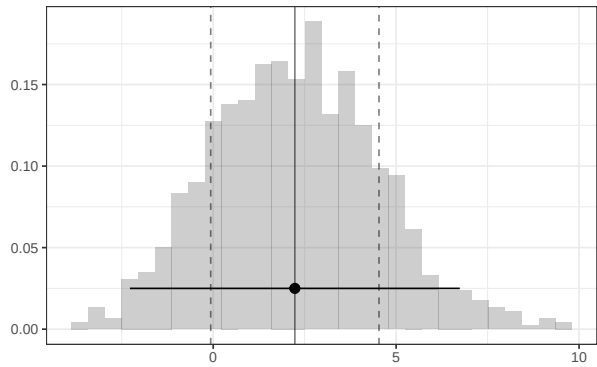


Figure 1: Histogram of the vector Y's elements

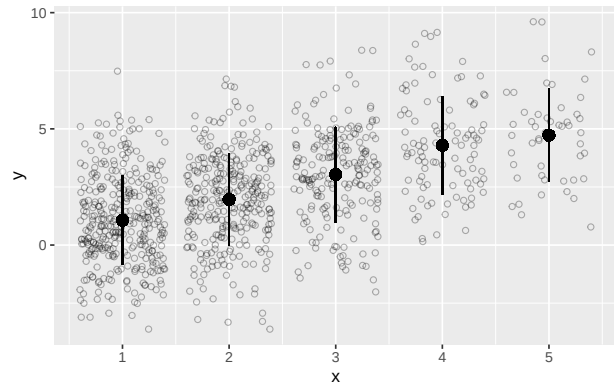


Figure 2: Jittered Scatterplot of X and Y

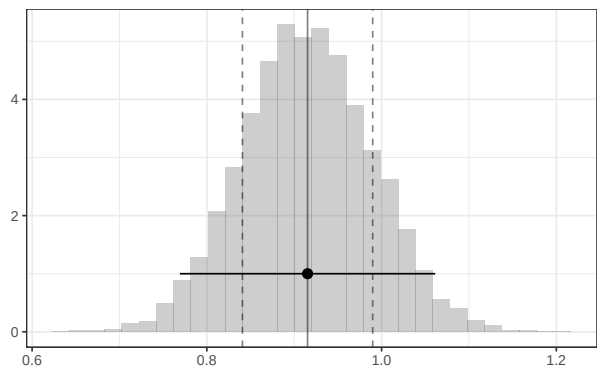


Figure 3: Histogram of the theta's elements

Part a

What does this code calculate? State it in words and mark it with an 'a', and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
mean(Y)
```

The sample mean (unconditioned). It's indicated by the solid vertical line on the sample histogram.

Part b

What does this code calculate? State it in words and mark it with a ‘b’, and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
sd(Y)
```

The sample standard deviation. It’s the distance between the solid vertical and either dashed vertical line on the sample histogram.

Part c

What does this code calculate? State it in words and mark it with a ‘c’, and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
mean(Y[X==1])
```

The conditional sample mean of Y among X=1. It’s indicated by the black dot on the scatterplot where X=1.

Part d

What does this code calculate? State it in words and mark it with a ‘d’, and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
sd(theta)
```

The bootstrap estimated sampling distribution’s standard deviation for the average increment estimator. Remember, theta is all our estimates of the average ‘over years’ across all our different samples. Theta gives us the estimated sampling distribution for our average increment estimator. It’s the distance between the solid vertical and either dashed vertical line on the sampling distribution histogram.

Part e

What does this code calculate? State it in words and mark it with a ‘e’, and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
(1/4)(mean(Y[X=5]) - mean(Y[X=1]))  
- 1.96* sqrt((1/4)^2*var(Y[X=5])/length(Y[X=5]) + (-1/4)^2*var(Y[X=1])/length(Y[X=1]))  
  
(1/4)(mean(Y[X=5]) - mean(Y[X=1]))  
+ 1.96* sqrt((1/4)^2*var(Y[X=5])/length(Y[X=5]) + (-1/4)^2*var(Y[X=1])/length(Y[X=1]))
```

A 95% confidence interval — based on normal approximation — for the average incremental difference in Y over X (from $X=1$ to $X=5$) in the population from which our data was drawn. This is indicated by the horizontal segment (with a dot in the middle) on the sampling distribution histogram (theta).

How do we calculate confidence intervals in general? For any point estimate, we might want to know what our population looks like— what population quantities would be consistent with our estimate? We can do this for any estimator— we just need to use the appropriate standard error (which we have to estimate) that corresponds to that estimator.

Mean $\pm 1.96 * est(SE_{Mean})$

Difference in Means $\pm 1.96 * est(SE_{Diff})$

Avg Increment Diff $\pm 1.96 * est(SE_{AvgIncrDiff})$

Weighted Avg Increment Diff $\pm 1.96 * est(SE_{WeightedAvgIncrDiff})$

$\hat{\beta} \pm 1.96 * est(SE_{\beta})$

Every estimator has its own variance and its own formula (if there is one) to estimate standard error.

Sometimes there is not a good formula for estimated standard error for certain estimators, or it is really complicated, or it relies on certain assumptions. So instead of worrying about deriving the variance of the estimator and figuring out a formula, we can use the estimated sampling distribution of our estimator to estimate our standard error. Remember, standard error is the standard deviation of the sampling distribution. So....

Part f

How could you calculate roughly the same thing as in part e using 'sd(theta)' replacing

$$\sqrt{((1/4)^2 \text{var}(Y[X=5]) / \text{length}(Y[X=5]) + (-1/4)^2 \text{var}(Y[X=1]) / \text{length}(Y[X=1]))}$$

We can use the standard deviation of the bootstrap estimated sampling distribution as our estimate of standard error.

$$(1/4)(\text{mean}(Y[X=5]) - \text{mean}(Y[X=1])) - 1.96 * \text{sd}(\text{theta})$$

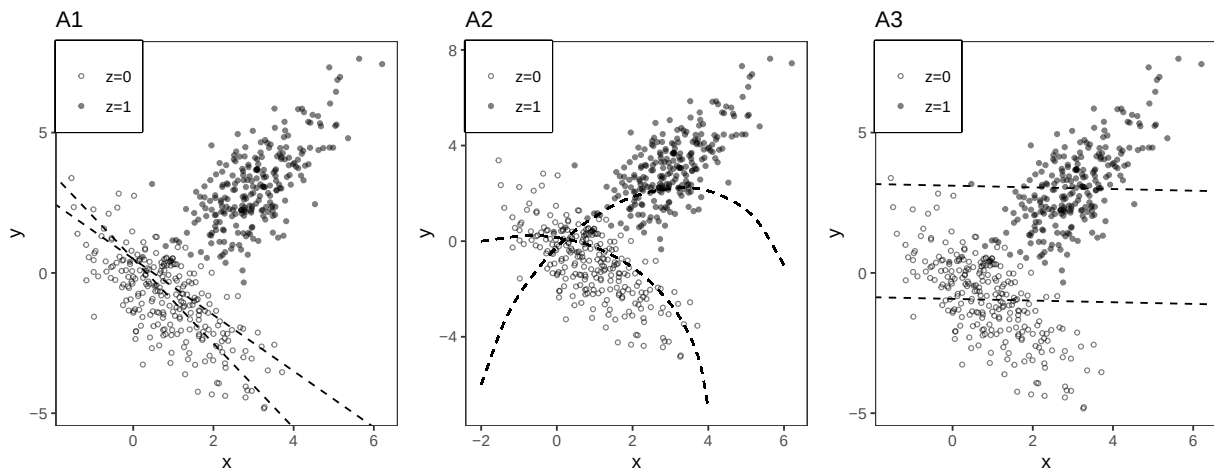
$$(1/4)(\text{mean}(Y[X=5]) - \text{mean}(Y[X=1])) + 1.96 * \text{sd}(\text{theta})$$

Problem 2

In the following parts, choose the figure with the appropriate regression line(s).

Part A

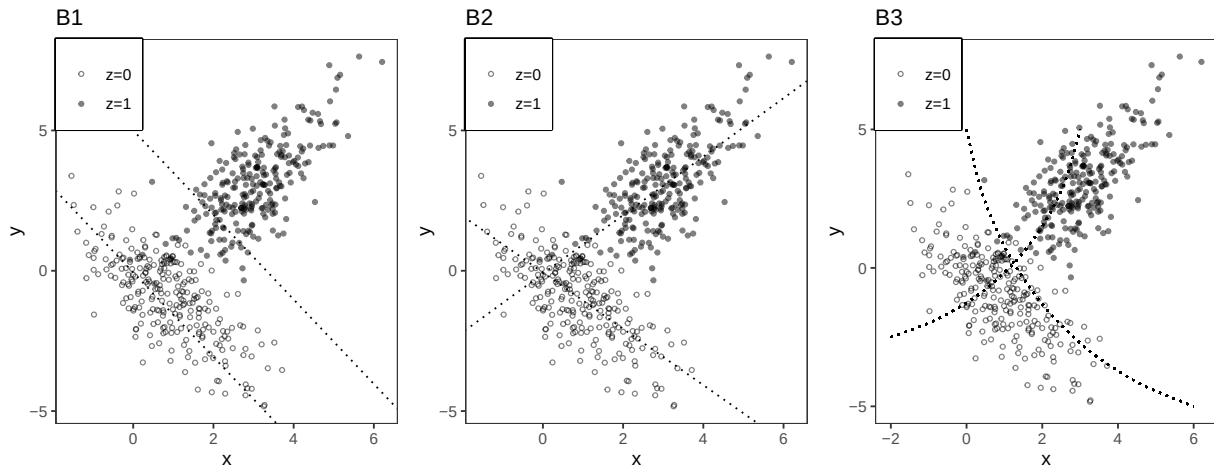
Choose the figure with the two dashed lines associated with the additive multiple regression ($y \sim x + z$).



A3

Part B

Choose the figure with the two dotted lines associated with the interactive multiple regression ($y \sim x * z$).



B2

Problem 3

Part A

We have collected y , x , and a binary variable z . The scatter plots below show our data.

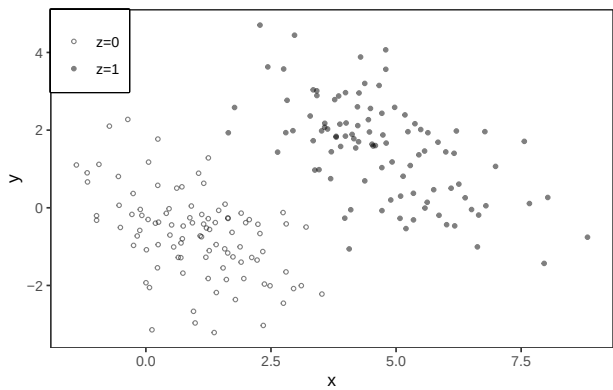


Figure 4: Scatterplot 1

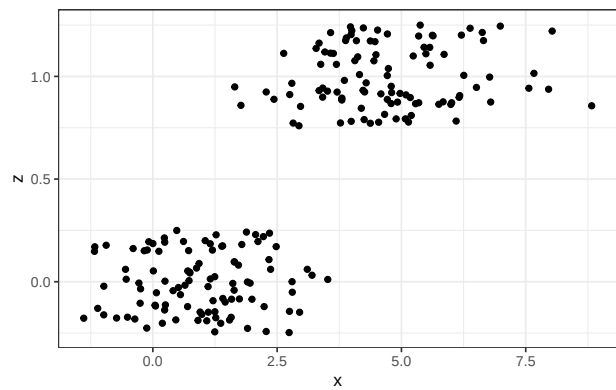


Figure 5: Scatterplot 2 (jittered)

Part A

Do you expect the coefficient on x from the simple regression of y on x ($lm(y \sim x)$) to be: (choose one, no explanation needed)

- Positive
- Negative
- Zero

Part B

We then run the regression $lm(y \sim x + z)$ which gives us the following output:

Call:

```
lm(formula = y ~ x + z, data = dfp3)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-2.9982	-0.6313	-0.0448	0.7168	3.2627

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.07683	0.11938	-0.644	0.521
x	-0.55172	0.05963	-9.253	<2e-16 ***
z	4.21009	0.26966	15.613	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.05 on 197 degrees of freedom

Multiple R-squared: 0.5961, Adjusted R-squared: 0.592

F-statistic: 145.4 on 2 and 197 DF, p-value: < 2.2e-16

And we run the regression $lm(z \sim x)$ which gives us the following output:

Call:

```
lm(formula = z ~ x, data = dfp3)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.62555	-0.20421	-0.00143	0.20566	0.71973

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.023784	0.031416	-0.757	0.45
x	0.184544	0.008657	21.317	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2768 on 198 degrees of freedom

Multiple R-squared: 0.6965, Adjusted R-squared: 0.695

F-statistic: 454.4 on 1 and 198 DF, p-value: < 2.2e-16

How would you use these outputs to calculate the coefficient on x from the simple regression $lm(y \sim x)$? Explain your steps and define any notation you use. Plug in the estimates from your output (you can leave it as an equation).

Let the simple regression of y on x be $y = \beta_0^S + \beta_1^S x$

Let the multiple regression of y on x and z be $y = \beta_0 + \beta_1 x + \beta_2 z$

Let the regression of z on x be $z = \gamma_0 + \gamma_1 x$ where $\gamma_1 = \frac{\sum_{i=1}^n (x_i - \bar{x}) z_i}{\sum_{i=1}^n (x_i - \bar{x})^2}$

From our OVB formula, we know $\beta_1^S = \beta_1 + \beta_2 * \gamma_1$

The regression outputs indicate that $\beta_1 = -0.5517$, $\beta_2 = 4.21$, and $\gamma_1 = 0.1845$

$\beta_1^S = -0.5517 + (4.21) * (0.1845) = 0.225$

Part C

Suppose we want β_1 from $y_i = \beta_0 + \beta_1 x_i + \beta_2 z_i + \beta_3 w_i + e_i$ but we run the regression $lm(y \sim x)$ without z or w . We have de-meaned our variables x_i , z_i , and w_i such that $\bar{x} = \sum_{i=1}^n x_i = 0$, $\bar{z} = \sum_{i=1}^n z_i = 0$, and $\bar{w} = \sum_{i=1}^n w_i = 0$. Under what condition(s) will the coefficient on x from $lm(y \sim x)$, which is $\frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2}$, be equal to β_1 ? Derive what your coefficient on x will be in terms of β s and justify your steps.

Side bar: de-meaned??? In lecture, we showed that demeaning a variable ensures that $\bar{x} = \sum_{i=1}^n x_i = 0$. Does that mean that $\sum_{i=1}^n x_i z_i = 0$? **No.** Let's convince ourselves.

	x^*	demeaned x	z^*	demeaned z	xz
	10		4		
	10		6		
	5		-2		
	0		2		
	0		0		
Avg.					
Sum					

The coefficient on x from $lm(y \sim x)$ is

$$\frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2}$$

Plug in y_i defined by the multivariate regression: $y_i = \beta_0 + \beta_1 x_i + \beta_2 z_i + \beta_3 w_i + e_i$

$$\frac{\sum_{i=1}^n x_i (\beta_0 + \beta_1 x_i + \beta_2 z_i + \beta_3 w_i + e_i)}{\sum_{i=1}^n x_i^2}$$

Distribute

$$\frac{\sum_{i=1}^n x_i \beta_0}{\sum_{i=1}^n x_i^2} + \frac{\sum_{i=1}^n x_i \beta_1 x_i}{\sum_{i=1}^n x_i^2} + \frac{\sum_{i=1}^n x_i \beta_2 z_i}{\sum_{i=1}^n x_i^2} + \frac{\sum_{i=1}^n x_i \beta_3 w_i}{\sum_{i=1}^n x_i^2} + \frac{\sum_{i=1}^n x_i e_i}{\sum_{i=1}^n x_i^2}$$

Let's work through each term.

$$\frac{\sum_{i=1}^n x_i \beta_0}{\sum_{i=1}^n x_i^2} = \frac{\beta_0 \sum_{i=1}^n x_i}{\sum_{i=1}^n x_i^2} = \frac{\beta_0 * 0}{\sum_{i=1}^n x_i^2} = 0$$

as x is demeaned, $\sum x_i = 0$

$$\frac{\sum_{i=1}^n x_i \beta_1 x_i}{\sum_{i=1}^n x_i^2} = \frac{\beta_1 \sum_{i=1}^n x_i x_i}{\sum_{i=1}^n x_i^2} = \beta_1$$

$$\frac{\sum_{i=1}^n x_i \beta_2 z_i}{\sum_{i=1}^n x_i^2} = \frac{\beta_2 \sum_{i=1}^n x_i z_i}{\sum_{i=1}^n x_i^2}$$

$$\frac{\sum_{i=1}^n x_i \beta_3 w_i}{\sum_{i=1}^n x_i^2} = \frac{\beta_3 \sum_{i=1}^n x_i w_i}{\sum_{i=1}^n x_i^2}$$

$$\frac{\sum_{i=1}^n x_i e_i}{\sum_{i=1}^n x_i^2} = \frac{0}{\sum_{i=1}^n x_i^2} = 0$$

we know that $\sum_{i=1}^n x_i e_i = 0$ from the first order conditions on our least squares estimator (we chose the intercept and coefficient(s) that minimize the sum of squared residuals)

Thus we have

$$\frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2} = \beta_1 + \frac{\beta_2 \sum_{i=1}^n x_i z_i}{\sum_{i=1}^n x_i^2} + \frac{\beta_3 \sum_{i=1}^n x_i w_i}{\sum_{i=1}^n x_i^2}$$

The coefficient on x from the simple regression of y on x will be equal to β_1 if: (1) β_2 or the coefficient on x from the simple regression from z on x is 0 and (2) β_3 or the coefficient on x from the simple regression of w on x is 0. If either of these terms is non-zero, our simple regression will not give us the same result as our regression that includes both x , z , and w . If x is randomized (like a random treatment assignment), it should not be correlated with z or w , so our short regression coefficient should give us β_1 .

Do we **need** our variables to be demeaned to derive that $\frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2} = \beta_1 + \frac{\beta_2 \sum_{i=1}^n x_i z_i}{\sum_{i=1}^n x_i^2} + \frac{\beta_3 \sum_{i=1}^n x_i w_i}{\sum_{i=1}^n x_i^2}$?

No! We did it in lecture without demeaning, it just took more steps to show that $\frac{\beta_0 \sum_{i=1}^n (x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2} = 0$

Problem 4

Consider the following R code:

```
1. lm(y ~ residuals(lm(z~x)))
2. lm(y ~ residuals(lm(x~z)))
3. lm(residuals(lm(y~z)) ~ x)
4. lm(residuals(lm(y~x)) ~ z)
5. lm(residuals(lm(y~x)) ~ residuals(lm(z~x)))
6. lm(residuals(lm(y~x)) ~ residuals(lm(x~z)))
7. lm(residuals(lm(y~z)) ~ residuals(lm(x~z)))
8. lm(residuals(lm(y~z)) ~ residuals(lm(z~x)))
```

- a) Choose one of the above that will produce the coefficient on x from the multiple regression of $y \sim x + z$

2 or 7

- b) BONUS: Choose one of the above that will produce the coefficient on z from the multiple regression of $y \sim x + z$

5 or 1

- c) Choose one of the above that will produce the same residuals as the residuals from the multiple regression of $y \sim x + z$

7 or 5